

Problem 4

$$y(n) = x(n) - \frac{1}{3} (x(n))^3 + \frac{1}{5} (x(n))^5$$

trig sub

$$y = \sin \theta - \frac{1}{3} (\sin \theta)^3 + \frac{1}{5} (\sin \theta)^5$$

$$y = \sin \theta + \frac{1}{3} \left(\frac{3 \sin \theta - \sin 3\theta}{4} \right) + \frac{1}{5} \left(\frac{10 \sin \theta - 5 \sin 3\theta + \sin 5\theta}{16} \right)$$

$$y = \frac{\sin \theta}{12} - \frac{3 \sin \theta}{12} + \frac{\sin 3\theta}{12} + \frac{10 \sin \theta}{80} - \frac{5 \sin 3\theta}{80} + \frac{\sin 5\theta}{80}$$

$$y = \frac{18 \sin \theta}{18} - \frac{2 \sin \theta}{18} + \frac{\sin \theta}{8} + \frac{\sin 3\theta}{12} - \frac{\sin 3\theta}{16} + \frac{\sin 5\theta}{80}$$

$$y = \frac{7 \sin \theta}{8} + \frac{16 \sin 3\theta}{192} - \frac{12 \sin 3\theta}{192} + \frac{\sin 5\theta}{80}$$

$$y = \frac{7 \sin \theta}{8} + \frac{(1) \sin 3\theta}{48} + \frac{\sin 5\theta}{80}$$